mhs-poisson-ltau: pressure-only magnetohydrostatic correction (τ grid)

Physics of `run.mode = "mhs-poisson-ltau"` (HSEMHS). 24 Oct 2026.

1. What the mode does

An inversion delivers T and $\mathbf{B}$ on surfaces of constant $\log_{10}\tau_{500}$. To use them physically we need the gas pressure, the density and the **geometric height** of every τ level, i.e. where in space the surfaces lie. `column-hse-ltau` answers this column by column. This mode adds the horizontal force balance between columns and the Lorentz force of the given field: with $T(\tau)$ and $\mathbf{B}(\tau)$ fixed, it finds $\ln P(\tau)$ and the heights $z(\tau)$ that minimise the force imbalance, in one non-linear least-squares solve directly on the τ grid. It is the optical-depth counterpart of `mhs-poisson-z`, and the mode intended for inversion output.

2. Quantities

Symbol	Meaning	Unit
τ	optical depth at 500 nm; levels uniform in $\log_{10}\tau$	–
T, $\mathbf{B}$	temperature and field on the τ levels, fixed	K, T
P, u	gas pressure on the τ levels; $u = \ln P$	Pa, –
$z(\tau)$	height of a level in a column; follows from P (Sect. 3)	m
$c(x, y)$	height of the cut level τ_c in a column (one unknown per column)	m
χ, ρ	extinction and density from the EOS, functions of (T, P)	m^{-1}, $\mathrm{kg\,m}^{-3}$
$\mathbf{L}$	Lorentz force $\mathbf{J}\times\mathbf{B}$, $\mathbf{J} = \nabla\times\mathbf{B}/\mu_0$ in physical space	$\mathrm{N\,m}^{-3}$
τ_c	cut: levels with $\log_{10}\tau \geq \log_{10}\tau_c$ are solved (production -4, $z \approx 0.56$ Mm)	–

3. Heights follow from the pressure

On a τ grid the geometry is not given: the spacing of two levels is $dz = -d\tau/\chi$, and χ depends on P. So the heights are part of the solution. In each column

$$z_k = c + \sum_{m=k_c}^{k-1} \Delta z_m, \qquad \Delta z_m = -\int_{\ln\tau_m}^{\ln\tau_{m+1}} \frac{\tau}{\chi(T,P)}\, d\ln\tau,$$

i.e. the height of every solved level is the height c of the cut level plus the thicknesses of the levels above it, each set by its own opacity. A change of pressure at one level shifts every level below it. Above the cut the column is kept hydrostatic in τ (as in `column-hse-ltau`) and moves rigidly with c.

4. The force balance in τ coordinates

The static balance $\mathbf{F} = -\nabla P + \mathbf{L} - \rho g\hat{\mathbf{z}} = 0$ must hold in physical space, but our derivatives are along the τ surfaces, which are tilted. For any quantity, $\partial_x f|_z = \partial_x f|_\tau - \partial_x z|_\tau\, \partial_z f$. Using the vertical balance $\partial_z P = -(\rho g - L_z)$:

$$\boxed{F_x = -\partial_x P\big|_\tau - (\rho g - L_z)\, \partial_x z\big|_\tau + L_x, \qquad \frac{d\ln P}{d\ln\tau} = \frac{(\rho g - L_z)\,\tau}{\chi\, P}}$$

(and the same for y). The horizontal equation is the familiar one with a geometric term: on a tilted τ surface, part of the weight of the gas acts sideways. The vertical equation is hydrostatic equilibrium in τ with the vertical Lorentz force reducing the effective weight. Both are divided by P (a relative imbalance, as in `mhs-poisson-z`), and the objective

$$\Phi = \sum_{\text{levels}} w\, \bar{H}^2 (R_x^2 + R_y^2) + \sum_{\text{intervals}} \frac{\bar{H}^2}{\Delta\bar{z}} R_z^2$$

weights each level by its thickness and median scale height, so $R\bar{H} \approx F/(\rho g)$ layer by layer.

$\mathbf{L}$ is computed from $\mathbf{B}$ on the τ surfaces at their current heights. When the heights change, the vertical spacing of the field samples changes and so do the currents; the solver follows this (Sect. 6 of the technical document). The field values themselves are never changed.

5. What decides the heights

Each column's heights relative to its neighbours are set by the horizontal balance: a column whose gas is lighter at a given height must be compressed (higher pressure) or displaced until the sideways pressure push vanishes. In

a strong-field column, the magnetic pressure carries part of the load, so less gas pressure is needed, the gas is less dense and less opaque, and $\tau = 1$ lies deeper: the **Wilson depression** emerges from the force balance, not from an assumption. One global constant (all columns moved together) is not fixed by the equations; it is set by requiring the mean height of $\tau = 1$ to be zero.

6. Results on the production cube

`IO/input-mhs-ltau.nc`, $504 \times 504 \times 95$ levels; 17 Gauss–Newton steps, 14 min on one node (8.3 min on four). Force imbalance after the solve, RMS over log-τ bands in units of ρg: horizontal 0.04–0.09 and vertical 0.04–0.06 for $\log\tau < 0$, 0.17 / 0.06 below $\tau = 1$ (start: 0.3–0.8). $z(\tau = 1)$ relative to the cube mean, median [5 %, 95 %] in km, binned by $|\mathbf{B}|$ at $\tau = 1$:

$\lvert\mathbf{B}\rvert$ at $\tau = 1$	column HSE in τ	fixed-point scheme (superseded)	**mhs-poisson-ltau**
0–200 G (245 123 columns)	−38 [−85, 187]	−55 [−257, 380]	−10 [−35, 73]
200–500 G	−68 [−88, 58]	+120 [−180, 549]	−26 [−52, 24]
500–1000 G	−86 [−104, −36]	+183 [−198, 587]	−51 [−89, −17]
1000–1500 G	−121 [−144, −92]	+314 [−117, 563]	−96 [−125, −63]
1500–3000 G (440)	−156 [−175, −140]	+402 [58, 568]	−130 [−167, −104]
columns outside −200 … + 300 km	2290	69 522	77

- The depression grows monotonically with field strength, somewhat less than in independent columns, because the neighbours now share the load through the horizontal balance.
- In weak-field regions the column-to-column scatter of $\tau = 1$ shrinks by a factor of about 3: horizontal pressure balance couples the columns that column HSE treats separately.
- An earlier scheme alternated a height-grid Poisson solve with re-integration of the τ columns. It lifted the strong-field columns instead of depressing them: it integrated the heights with a pressure that it then replaced. It is kept only as history (`scripts/poisson/poisson_tau.jl`).

7. Limitations

- As in `mhs-poisson-z`, a fixed $\mathbf{B}$ cannot balance the buoyancy of the temperature structure; the remaining imbalance is of that kind. Above the cut ($\log\tau < -4$) the columns are only hydrostatic.
- The heights of a few strong-field columns are only weakly constrained: when the solve stops, they still move by a few km per step while Φ changes by less than 10^{-4}. Different runs agree to about 1 m at the median and 12–24 m at the 99th percentile.
- The vertical integration uses the trapezoid rule on the 0.1 dex grid; sharp temperature jumps on that grid make individual level thicknesses uncertain (see the cap in `column-hse-ltau`).